

Hybrid Biomedical Image-Analysis Pipeline: Fluorescence Microscopy Nuclei Segmentation

AI Imaging Coding Case Study and Report

1. Introduction

This report presents a hybrid biomedical image-analysis pipeline for the assigned modality — fluorescence microscopy nuclei (a synthetic 112-image dataset, in the spirit of the 2018 Data Science Bowl nuclei set, with exact ground-truth binary and instance masks). The pipeline compares a multimodal LLM's direct image description against a classical-features-plus-LLM numbers-first description, trains and evaluates a U-Net, and assembles the whole system into an auditable per-image record: raw image → segmentation → quantitative region features → structured JSON record → narrative.

2. Data Preparation and Multimodal LLM Description

Images were converted to grayscale and confirmed at the common 256×256 size (already native resolution). Figure 1 shows four representative training images and the pooled pixel-intensity histogram across all 80 training images, which is clearly bimodal — a background spike near 0 and a foreground bump around 0.2–0.35 — justifying Otsu thresholding as a reasonable classical baseline (Part 3).

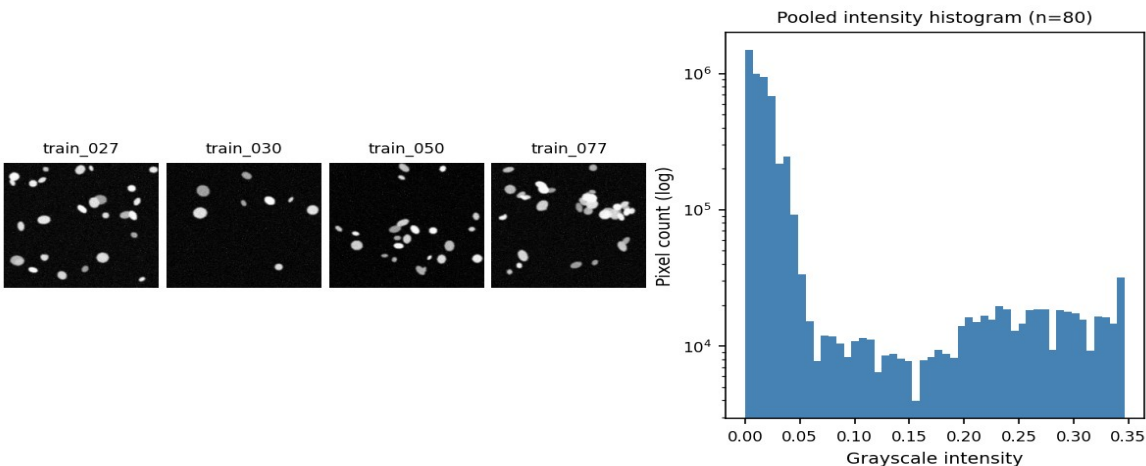


Figure 1. Sample training images (top) and pooled intensity histogram (bottom-right).

The representative image train_062 (27 ground-truth nuclei, “normal” density, the regime's median — not cherry-picked) was sent to llama3.2-vision under two prompts. A naive prompt (“What is in this image?”) produced a detailed but diagnostically-drifted response: it correctly identified DAPI staining, but confidently claimed the cells were “actively dividing or have recently divided” and in a “state of rapid growth or proliferation” — unverifiable claims from a single static image. The optimised prompt below anchored the model as descriptive rather than diagnostic, forced a fixed JSON schema, and explicitly permitted “uncertain.” Its response correctly identified modality as “fluorescence,” honestly flagged tissue_type as “uncertain,” and stayed strictly observational.

Optimised prompt (Task 1):
You are a scientific image-cataloguing assistant helping annotate microscopy images for a research dataset. You are NOT a diagnostic tool. Do not provide any clinical, diagnostic, or disease-related interpretation, even if asked. Describe only what is visually present, in purely observational terms (shapes, brightness, spatial arrangement, apparent density, texture).
Respond with a JSON object with EXACTLY these four fields, and nothing else:
{ "modality": "<...or 'uncertain'>", "tissue_type": "<...or 'uncertain'>",
"notable_features": "<one short factual sentence>", "image_quality": "<good|moderate|poor>" }
If you are not confident about a field, write "uncertain" rather than guessing. Respond with ONLY the JSON object — no preamble, no explanation, no markdown code fences.

Three repeated calls at temperature=0.8 on the same prompt and image were NOT identical: modality confidence flip-flopped (“fluorescence microscopy” → “fluorescence” → “uncertain”) and image_quality flip-flopped (“good,” “good,” “poor”) on the same static image — genuine run-to-run instability even under a well-constrained prompt.

3. Classical Features and LLM Interpretation

Otsu thresholding (with morphological cleanup and small-object removal) plus connected-component labeling detected 22 objects in train_062 (27 ground-truth nuclei; some touching nuclei merged into single blobs). A regionprops feature table (area, perimeter, eccentricity, solidity, mean intensity, extent) was converted into a deterministic text summary and passed — numbers only, never the image — to llama3.2.

Numbers-first prompt (Task 2, template):
You are a scientific image-cataloguing assistant. You have NOT seen the image itself — you are working ONLY from a numeric summary produced by classical image segmentation (Otsu thresholding). Do not claim

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to have observed anything beyond what the summary states, and do not provide any clinical or diagnostic
interpretation. Image ID: <id> Summary: <regionprops-derived text summary>
Based ONLY on this summary, produce a JSON object with EXACTLY: n_objects, density_class (sparse/
moderate/dense), shape_regularity (highly irregular/irregular/regular/highly regular), quality_flag
(ok/review_recommended/fail); then a one-paragraph description grounded strictly in the summary's numbers.
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The response correctly echoed `n_objects=22` (a direct pass-through) but misclassified `density_class` as “dense” (ground truth: normal), since the model had no dataset-wide reference for what “dense” means from one image’s numbers alone. `shape_regularity=“irregular”` is also questionable: solidity (0.95, near-maximal convexity) argues for regular boundaries while eccentricity (0.69) mildly argues for elongation — the model picked one without acknowledging the tension. The narrative also mis-explained “extent” (area/bounding-box area, a compactness measure) as “indicating a mix of small and large objects” — a reasoning error layered on a correctly-cited number, not a fabricated one.

4. U-Net Segmentation

The provided small U-Net (3-level encoder+bottleneck, base=16 channels, 483,153 parameters, matching the course lab skeleton) was trained for 10 epochs with combined BCE+Dice loss and Adam ($\text{lr}=1\text{e-}3$) on the real dataset (80 train / 20 val), replacing only the lab’s synthetic data generator with the real images. Training and validation loss decreased smoothly with no overfitting; final validation Dice = 0.9828, IoU = 0.9662 — exceeding the lab’s own stated benchmark range (0.85–0.95).

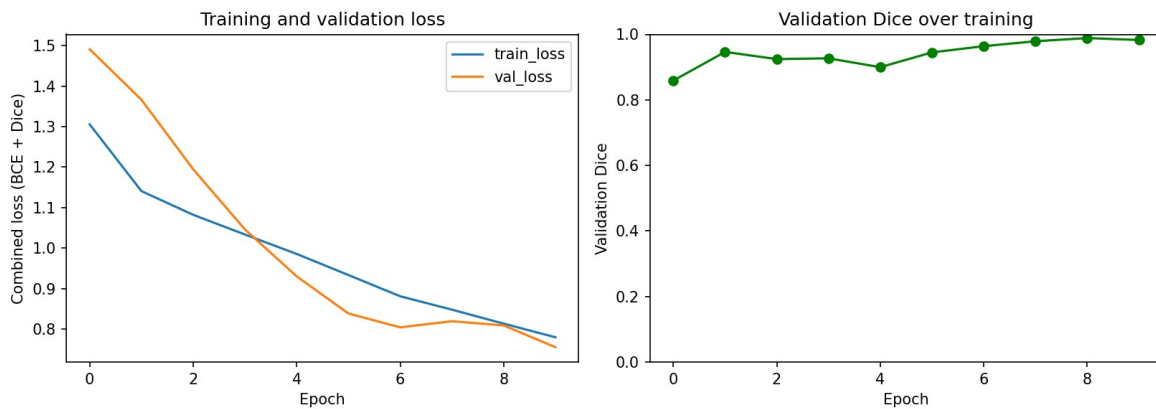


Figure 2. Training/validation loss and validation Dice across 10 epochs.

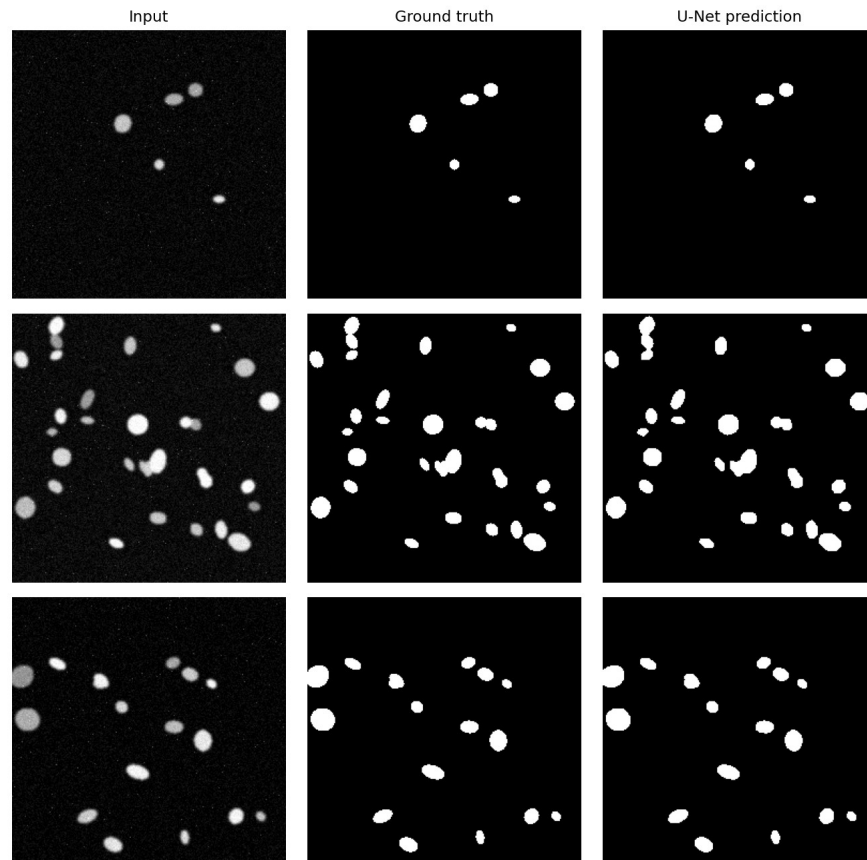


Figure 3. Input, ground truth, and U-Net prediction for three validation images.

Comparing against Otsu directly on the 12 unseen test images gives a genuinely two-part answer (Table 1). Object counts (connected-component labeling) were essentially tied (mean absolute error: Otsu 12.25, U-Net 12.58) — not a U-Net failure, but because the ground-truth BINARY masks themselves merge touching nuclei (only the separate 16-bit instance labels distinguish them; the U-Net trained on binary masks inherits the same limitation). Pixel-level Dice — the metric each model actually optimises — showed a real, modest U-Net advantage: mean Dice 0.9815 (U-Net) vs. 0.9784 (Otsu), ahead on 10/12 images. Largest margin: test_005 (clustered, 49 nuclei; Otsu 0.978, U-Net 0.989). Otsu's best relative case: test_003 (normal, 21 nuclei; Otsu 0.981, U-Net 0.976).

Metric	Otsu	U-Net
Mean pixel Dice (12 test images)	0.9784	0.9815
Mean pixel IoU (12 test images)	0.9577	0.9638
Mean abs. object-count error	12.25	12.58
Validation Dice / IoU (20 val images)	—	0.9828 / 0.9662

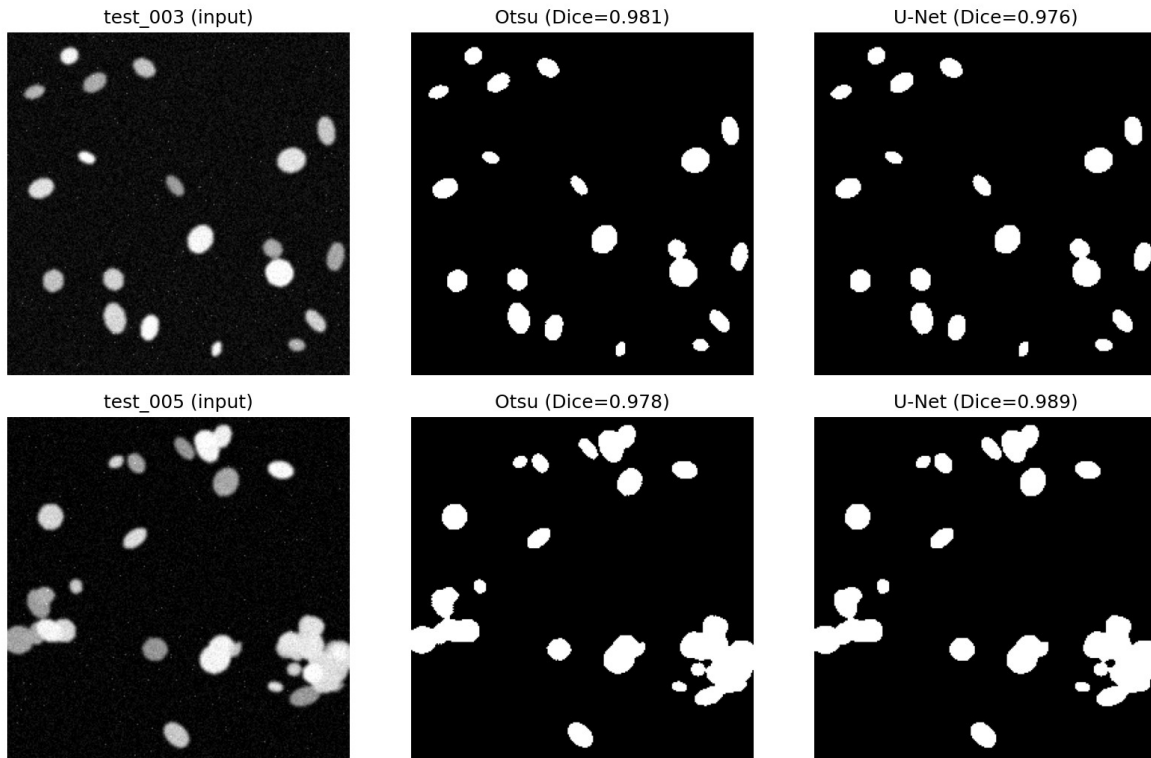


Figure 4. Best Otsu case (test_003, top) vs. largest U-Net advantage (test_005, bottom).

5. Hybrid Pipeline

The trained U-Net segmented all 12 unseen test images; each mask was converted to a regionprops feature table, summarised, and passed to llama3.2 to produce a five-field JSON record and narrative, aggregated into a CSV across all 12 images.

Hybrid prompt (Task 4, template):

You are a scientific image-cataloguing assistant... You have NOT seen the image itself — you are working ONLY from a numeric summary produced by U-Net segmentation. [Same descriptive-not-diagnostic anchoring and numbers-only framing as Task 2.] Produce a JSON object with EXACTLY: image_id, n_objects, mean_area, density_class (sparse/moderate/dense), quality_flag (ok/review_recommended/fail); then a one-paragraph narrative grounded strictly in the summary.

A striking finding: density_class was “moderate” for every single one of the 12 images, despite n_objects ranging from 8 to 42 (a >5× spread) — while n_objects and mean_area (direct numeric pass-throughs) varied correctly every time. quality_flag was “ok” throughout too, but this appears legitimately correct rather than miscalibrated, since none of these 12 images were actually corrupted (unlike the separate test_corrupted set). This is the clearest evidence in the whole pipeline that hallucination risk concentrates in unanchored categorical judgments, not in directly-extracted numbers — directly informing Q4 below.

image_id	n_objects	mean_area	density_class	quality_flag
test_000	8	197	moderate	ok
test_004	42	345	moderate	ok
test_010	35	388	moderate	ok

Table 2. Example rows from the aggregated CSV (full 12-row table in outputs/metrics/task4_hybrid_results.csv), showing correct numeric pass-through alongside uniformly uninformative density_class.

Required Questions

Q1. Which description is more useful/trustworthy — VLM (Task 1) or numbers-first (Task 2)?

Usefulness and trustworthiness split. The optimised VLM description is more useful — it captures qualitative visual character a numbers-only summary cannot — but only when constrained: the naive prompt fabricated unverifiable biological claims (“actively dividing”). Numbers-first is more trustworthy only for direct pass-through fields (`n_objects`); its categorical judgment (`density_class`) was wrong. The key differentiator is auditability: numeric fields check against the `regionprops` table directly; a VLM’s claim about biological state cannot be checked against anything.

Q2. Did the U-Net improve on Otsu? One example each did better.

Two-part answer. Object counting: no improvement, tied (12.25 vs. 12.58 error) — both share the same binary-mask instance-merging limitation, not a modelling failure (matches the course lecture’s watershed/instance-segmentation guidance). Pixel Dice (the metric each model optimises): real U-Net advantage, 10/12 images, mean 0.9815 vs. 0.9784. U-Net better: `test_005` (clustered; Otsu 0.978, U-Net 0.989). Otsu better: `test_003` (normal; Otsu 0.981, U-Net 0.976), margin small.

Q3. Report Dice/IoU. What do they mean, and where does the model make mistakes?

Dice=0.9828, IoU=0.9662 (validation). Both measure pixel overlap (0–1); IoU is stricter ($\text{IoU} = \text{Dice} / (2 - \text{Dice})$). Errors don’t track density simply — sparse `test_006` is among the weaker cases while dense `test_004/010` score well — suggesting boundary precision matters more than scene complexity. Visual inspection: errors concentrate at object boundaries (CNN upsampling smooths angular edges), disproportionately affecting small objects. No formal per-region quantitative breakdown was computed — a natural extension beyond what was done here.

Q4. Where can the LLM hallucinate, what reduces the risk, and why does JSON-as-source-of-truth help?

Four modes found: (1) fabrication with no grounding (Task 1 naive: “actively dividing”); (2) run-to-run instability (Task 1 optimised, modality/quality flip-flopped across identical repeats); (3) miscalibrated categorical judgment (Task 2’s wrong `density_class`; Task 4’s uniform “moderate” despite `n_objects` 8–42); (4) incorrect reasoning about a correctly-cited fact (Task 2’s extent misinterpretation). Mitigations used: descriptive-not-diagnostic anchoring; explicit “uncertain” permission (under-used in practice); numbers-only input (turns unfalsifiable perception-hallucination into checkable reasoning-hallucination); fixed enumerated schema; temperature=0 for real outputs. JSON-as-source-of-truth helps because it separates auditable numeric fields from the categorical/narrative layer shown here to be unreliable — Task 4’s CSV is built from JSON, not parsed narratives, so downstream users inherit numeric reliability specifically, not narrative-layer hallucination.

Q5. Would you trust any part of this system clinically? What single change would help most?

No. Three compounding reasons: the dataset is fully synthetic and small, so no result here licenses any real-tissue claim; segmentation cannot separate touching nuclei into correct counts (errors up to 42 objects), which most clinical use cases would need; every LLM stage showed a directly demonstrated, reproducible failure in this project’s own evidence, matching the brief’s “not cleared for clinical use” framing concretely. Single most impactful change: retrain segmentation as instance-aware using the already-available 16-bit instance labels (or add marker-based watershed) — chosen over LLM-calibration fixes because the counting failure is the most severe, most load-bearing problem in the pipeline, feeding every downstream summary, LLM call, and the final CSV.

Discussion and Limitations

The dataset is fully synthetic (generated by `make_dataset.py`, CC0), chosen for reproducibility and an exact answer key — but this is itself a limitation for any real-world generalisation claim (Q5). Extension work (robustness tracing, loss ablation, foundation-model swap) was not pursued, prioritising depth on the five required tasks and questions within the page budget over additional breadth. Natural next steps beyond Q5’s single change: a formal per-region/object-size error breakdown for the U-Net (Q3), and giving the LLM explicit dataset-wide calibration context before requesting `density_class`, as a cheaper complementary fix to instance segmentation.

References

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